DST-FIST SPONSORED

TWO WEEKS INTERNSHIP AND HANDS-ON TRAINING WITH ANALYTICAL INSTRUMENTS

AN INDUSTRIAL TRAINING REPORT

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In partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

BIOTECHNOLOGY

RAJALAKSHMI ENGINEERING COLLEGE

(AN AUTONOMOUS INSTITUTION AFFILIATED TO ANNA UNIVERSITY)

THANDALAM, CHENNAI 602105

APRIL 2025

BONOFIDE CERTIFICATE

Certified that this training report titled "**TWO WEEKS INTERNSHIP AND HANDS-ON TRAINING WITH ANALYTICAL INSTRUMENTS**" is the Bonafide work of **VIJAY SURYA A V (2116220401022), BALAJI L (2116220401027), BHUBESHNATH P (2116220401030)** who carried out the internship under my supervision. Certified further that to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was confirmed on an earlier occasion on this or any other candidate.

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DST-FIST SPONSORED ANALYTICAL INSTRUMENTS
Two Weeks Internship & Hands-on Training

25 years OF ACADEMIC EXCELLENCE

Rajalakshmi Institutions

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This is to certify that

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of RAJALAKSHMI ENGINEERING COLLEGE

has participated & completed the **"Two weeks Internship and Hands-on training with Analytical Instruments"** organized by DST-FIST Sponsored Sophisticated Analytical Instruments Laboratory for Multidisciplinary Research in Biological Sciences in association with the **Department of Food Technology and Biotechnology** from 10.07.2024 to 20.07.2024 at **Rajalakshmi Engineering College**, Chennai. We wish him/her the very best of luck in future endeavors.

Mr. R. Anand Assistant Professor, Coordinator	**Dr. K. Ramalakshmi** Professor & Head, Implementation Agency Maker	**Dr. S.N. Murugesan** Principal Implementation Agency Checker

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DECLARATION

I hereby declare that the project entitled "**TWO WEEKS INTERNSHIP AND HANDS-ON TRAINING WITH ANALYTICAL INSTRUMENTS**" is a Bonafide work carried out by me under the supervision of **Dr. Rekha Ravindran**, Assistant Professor, Department of Biotechnology, Rajalakshmi Engineering College, Thandalam, Chennai (affiliated to Anna University, Chennai) and thank **Anand R**, Assistant professor, Department of Food Technology, Rajalakshmi Engineering College, Chennai (affiliated to Anna University, Chennai) towards the partial fulfilment of the degree of B.Tech., Biotechnology.

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TABLE OF CONETNTS

CHAPTER NO.	TITLE	PAGE NO.
	TABLE OF CONTENTS	I
	LIST OF FIGURES	II
	LIST OF ABBREVIATIONS	III
	LIST OF TABLES	IV
	ABSTRACT	1
1	INTRODUCTION	2
2	AIM AND OBJECTIVES	5
3	PRINCIPLE OF INSTRUMENTS	7
4	MATERIALS AND METHODS	10
5	RESULTS	21
6	CONCLUSION	28
7	REFERENCES	29

TABLE OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
1	Ultra-High Performance Liquid Chromatography	19
2	Plant Extraction and Nanoparticle from plant extract	20
3	SARS-CoV-2 in human respiratory specimens using RT-qPCR.	21
4	Stained Bacterial Cells	23
5	Live Cells	23
6	Dead Cells	24
7	Stained Bacterial Cells By DAPI Staining Method	24
8	Caffeine Standard and Unknown Sample	26
9	Absorbance peak at 352 nm	27

LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
RT-PCR	Reverse Transcription Polymerase Chain Reaction
cDNA	Complementary DNA
RNA	Ribonucleic Acid
DNA	Deoxyribonucleic Acid
UHPLC	Ultra-High Performance Liquid Chromatography
HPLC	High Performance Liquid Chromatography
UV-Vis	Ultraviolet-Visible Spectroscopy
A260/A280	Absorbance Ratio At 260 nm And 280 nm
A260/A230	Absorbance Ratio At 260 nm And 230 nm
NGS	Next-Generation Sequencing
AI	Artificial Intelligence
STED	Structured Illumination Microscopy
MS	Mass Spectroscopy
qRT-PCR	Quantitative Reverse Transcription Polymerase Chain Reaction

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
1	Fluorophore-Quencher Combination for SARS-CoV-2 RT-PCR Target.	10
2	Mixture of reagents for no. of reaction cycles	11
3	Steps in RT-PCR for detection.	11
4	Determination of Ct value for positive and negative results.	21
5	Amplification of Covid-19 genes.	22

ABSTRACT

This report presents the outcomes of a two-week intensive internship conducted at the DST-FIST sponsored Sophisticated Analytical Instruments Laboratory, Rajalakshmi Engineering College. The program provided in-depth, hands-on exposure to high-end analytical instruments widely employed in contemporary biological and biomedical research. Key instrumentation included Reverse Transcription Polymerase Chain Reaction (RT-PCR) for mRNA-to-cDNA conversion and amplification to assess gene expression profiles; Fluorescence Microscopy for high-resolution visualization of fluorophore-labeled biomolecules and subcellular structures; Ultra-High Performance Liquid Chromatography (UHPLC) for high-efficiency separation and quantification of bioactive compounds; and the NanoDrop Spectrophotometer for microvolume spectrophotometric analysis of nucleic acids and proteins, emphasizing concentration and purity assessment.

The training module encompassed theoretical instruction, standard operating procedures (SOPs), instrument calibration, method development, and research-oriented project work. The focus areas aligned with current research priorities, including molecular characterization of pathogens, quantitative analysis of microbial by-products, environmental bioremediation, nano-material synthesis, and nutraceutical evaluation. This technical exposure enhanced the participant's capabilities in experimental design, analytical precision, and data interpretation—key competencies in life science and biotechnology domains.

KEYWORDS:

RT-PCR, Fluorescence Microscopy, UHPLC, NanoDrop Spectrophotometry, Gene Expression, Molecular Diagnostics, Bioactive Compounds, Nano-materials, Analytical Techniques, Microbial Metabolites, Bioremediation

CHAPTER 1

INTRODUCTION

In the field of biological sciences, the integration of advanced analytical instrumentation has become indispensable for the accurate assessment of molecular, biochemical, and cellular processes. Techniques such as Reverse Transcription PCR (RT-PCR), Fluorescence Microscopy, Ultra-High Performance Liquid Chromatography (UHPLC), and Nano drop Spectrophotometry are widely applied in research involving gene expression, cellular imaging, compound analysis, and nucleic acid quantification [1], [2].

1.1. Reverse Transcription Polymerase Chain Reaction (RT-PCR)

RT-PCR is a sensitive and specific technique for detecting and quantifying gene expression by converting RNA into complementary DNA (cDNA) followed by amplification. This method is critical in virology, oncology, and transcriptomics for analyzing differential gene expression and pathogen detection [3], [4]. RT-PCR not only enables the detection of gene expression but also facilitates the examination of post-transcriptional modifications, splicing events, and RNA variants. The technique is widely employed in clinical diagnostics, particularly in the detection of viruses like SARS-CoV-2, where it provides essential information on viral load and infection stages [4]. RT-PCR is also pivotal in monitoring therapeutic responses in cancer research, as it allows for the evaluation of gene expression profiles associated with specific tumor types and the effectiveness of targeted treatments [3]. Furthermore, quantitative RT-PCR (qRT-PCR) has become a standard tool in molecular biology for assessing relative changes in gene expression across various conditions, making it indispensable for studies on gene regulation and cellular responses to environmental stimuli [4].

1.2.Fluorescence Microscopy

Fluorescence microscopy allows visualization of cellular structures and biomolecules using fluorophore-labeled probes. It is extensively used in cell biology and immunofluorescence assays to detect protein localization, intracellular signaling, and dynamic cellular processes [5]. Fluorescence microscopy has evolved significantly with advancements in super-resolution techniques, such as STED (Stimulated Emission Depletion) and SIM

(Structured Illumination Microscopy), which push the boundaries of spatial resolution beyond the diffraction limit of conventional optics [5]. These high-resolution techniques allow for the visualization of subcellular structures, including organelles, proteins, and nucleic acids, with unprecedented detail. Fluorescence microscopy is particularly valuable in live-cell imaging, enabling real-time observation of dynamic processes such as protein-protein interactions, signal transduction pathways, and cellular movements [5]. Moreover, advances in multi-plexing techniques have made it possible to track multiple biomolecules simultaneously, providing insights into the interplay of various cellular components [5]. Applications extend beyond basic research into clinical settings, where fluorescence microscopy aids in diagnostic pathology, cancer imaging, and early detection of disease biomarkers [5].

1.3.Ultra-High Performance Liquid Chromatography (UHPLC)

UHPLC is a high-resolution chromatographic technique offering rapid and precise separation of analytes in complex biological matrices. Its application spans pharmaceutical analysis, metabolomics, and environmental monitoring due to its increased sensitivity and speed compared to conventional HPLC [6], [7]. UHPLC has rapidly become a standard in analytical laboratories due to its ability to separate and quantify complex mixtures with high resolution and speed. Unlike traditional HPLC, UHPLC employs smaller particle sizes, which allows for higher pressure operation and faster analysis times, making it ideal for high-throughput environments [6]. In pharmacokinetics, UHPLC is invaluable for the rapid analysis of drug concentrations in biological fluids, ensuring that pharmacological studies are both time-efficient and precise [6]. The technique also plays a crucial role in metabolomics, where it helps identify and quantify metabolites in biological samples, contributing to our understanding of metabolic disorders, disease mechanisms, and therapeutic interventions [7]. In the food and environmental sectors, UHPLC is applied to detect contaminants, pesticides, and pollutants, thus ensuring safety and compliance with regulatory standards [6], [7].

1.4.Nano Drop Spectrophotometry

The Nano Drop spectrophotometer enables micro volume quantification of DNA, RNA, and proteins using UV-Vis absorbance, with high reproducibility and minimal sample volume. This method is routinely used for assessing sample purity (via A260/A280 and A260/A230 ratios) prior to downstream applications like PCR, sequencing, or electrophoresis [8]. The Nano Drop spectrophotometer revolutionized nucleic acid and protein quantification by requiring only small sample volumes (typically 1–2 μL), which is particularly advantageous when working with precious or limited samples [8]. Its high sensitivity and precision make it an essential tool in molecular biology workflows, particularly for RNA and DNA quantification prior to downstream applications such as sequencing, PCR, and microarray analysis [8]. The Nano Drop's ability to provide quick and accurate concentration and purity measurements (using the A260/A280 and A260/A230 ratios) significantly reduces the time and effort required for sample preparation and ensures the reliability of subsequent experiments [8]. Furthermore, the Nano Drop is an indispensable tool in proteomics, where it is used to quantify and assess the purity of protein samples before mass spectrometry or functional assays. The integration of this tool with automated sample handling systems further streamlines workflows, making it an indispensable piece of equipment in modern molecular biology laboratories [8].

CHAPTER 2

AIM AND OBJECTIVES

2.1.AIM

The aim of this industrial training is to build advanced technical competencies in the operation, application, and data interpretation of high-throughput analytical instrumentation—specifically RT-PCR, Fluorescence Microscopy, UHPLC, and NanoDrop Spectrophotometry—for use in molecular biology, analytical biochemistry, and cellular diagnostics.

2.2.OBJECTIVES

1. To perform RNA isolation, reverse transcription, and quantitative amplification using Reverse Transcription Polymerase Chain Reaction (RT-PCR) for the analysis of differential gene expression and pathogen detection in biological samples.

2. To visualize and characterize biomolecular structures and subcellular components through Fluorescence Microscopy by applying fluorochrome-based labeling, understanding excitation/emission spectra, and image acquisition via epifluorescence or filter-based optics.

3. To achieve efficient chromatographic separation and quantification of complex biological mixtures such as plant extracts, microbial metabolites, or pharmaceutical compounds using Ultra-High Performance Liquid Chromatography (UHPLC), including method development, gradient optimization, and peak integration.

4. To quantify nucleic acids and proteins with high accuracy and minimal sample volume using NanoDrop UV-Vis Spectrophotometry, including interpretation of A260/280 and A260/230 ratios for purity analysis and downstream compatibility in molecular workflows.

5. To adhere to analytical method validation criteria including linearity, precision, accuracy, sensitivity (LOD/LOQ), and reproducibility during experimental runs, aligned with ICH and GLP standards.

6. To implement standard operating procedures (SOPs) for instrument calibration, decontamination, and data logging, ensuring experimental consistency and compliance with biosafety regulations.

7. To develop competency in interpreting raw data outputs such as amplification curves (Ct values), fluorescence emission spectra, chromatograms, and absorbance spectra, translating them into biologically meaningful results.
8. To correlate instrumentation output with real-world biological applications, such as microbial diagnostics, nutraceutical profiling, bioavailability assessment, and nanomaterial characterization.

CHAPTER 3

PRINCIPLE OF INSTRUMENTS

3.1.REVERSE TRANSCRIPTASE-POLYMERASE CHAIN REACTION : COVID-19 ANALYSIS

CoviDxTM mPlex-4R SARS-CoV-2 RT-PCR Detection Kit is an in-vitro RT-PCR qualitative assay for the specific detection of SARS-CoV-2 in human respiratory specimens. CoviDxTM mPlex-4R SARS-CoV-2 RT-PCR is a multiplex assay kit where detection of four genes (three SARS-CoV-2 and one human internal control) can be detected in a single tube reaction. This CoviDxTM mPlex-4R SARS-CoV-2 primer and probe sets are designed for the detection of E-gene, N-gene, and RdRP-gene of SARS-CoV-2 along with human internal control RNase P in a single tube assay, based on the Berlin protocol.

3.2.FLUORESCENENCE MICROSCOPY EXPERIMENTS

3.2.1. CRYSTAL VIOLET BIOFILM FORMATION ASSAY

Biofilms are biological aggregates composed of bacteria and extracellular polysaccharides that can live in a variety of environments. Biofilms are responsible for bacterial contamination of medical devices as well as infectious disorders such as cavities and periodontitis. Biofilm microorganisms are very resistant to antibiotics. The Crystal Violet Biofilm Formation Assay assesses the quantity of biofilm formation and anti-biofilm activity of samples. Because the biofilm forms on a peg layer on the lid of a microplate, numerous samples can be cleaned and quantified simultaneously without having to peel off the biofilm for each activity.

3.2.2. DRUG TOXICITY USING FLUORESCENT MICROSCOPY

Bacterial Viability Kits provide a novel two-color fluorescence assay of bacterial viability that has proven useful for a diverse array of bacterial genera. This kit utilizes mixtures of our SYTO 9 green-fluorescent nucleic acid stain and the red-fluorescent nucleic acid stain, propidium iodide. These stains differ both in their spectral characteristics and in their ability to penetrate healthy bacterial cells. When used alone, the SYTO 9 stain generally labels all bacteria in a population — those with intact membranes and those with damaged membranes.

In contrast, propidium iodide penetrates only bacteria with damaged membranes,causing a reduction in the SYTO 9 stain fluorescence when both dyes are present. Thus, with an appropriate mixture of the SYTO 9 and propidium iodide stains, bacteria with intact cell membranes stain fluorescent green, whereas bacteria with damaged membranes stain fluorescent red. The excitation/emission maxima for these dyes are about 480/500 nm for SYTO 9 stain and 490/635 nm for propidium iodide.

3.2.3. ORGANELLES FIXATION AND VISUALIZATION

Fixation stabilizes cellular structures and organelles by preserving their morphology and preventing autolysis or degradation. Visualization involves staining or labeling organelles to make them distinguishable under a microscope. The process combines chemical fixation, permeabilization, and staining or labeling with specific dyes or antibodies, followed by imaging.

3.3.ULTRA-HIGH PERFORMANCE LIQUID CHROMATOGRAPHY : QUANTIFICATION OF CAFFEINE IN BEVERAGES

Ultra-High Performance Liquid Chromatography (UHPLC) is a highly efficient technique used for the rapid and precise quantification of caffeine in beverages. It operates on the principles of liquid chromatography, where a sample is injected into a column packed with a stationary phase, and compounds are separated based on their chemical properties. The mobile phase, typically a mixture of water and an organic solvent, facilitates this separation. Caffeine, which absorbs UV light at 273 nm, is detected using a UV-Vis detector, and its concentration is determined by comparing the peak area or height to a calibration curve created from known caffeine standards. The benefits of UHPLC, including higher resolution, faster analysis, and greater sensitivity, make it ideal for analyzing complex beverage matrices like soft drinks, energy drinks, coffee, and tea. However, challenges such as matrix effects and the need for proper calibration must be addressed for accurate results.

3.4.NANOPARTICLES (OR) NANOEMULSION SYNTHESIS

The NanoDrop spectrophotometer is a highly sensitive, small-volume UV-Vis spectrophotometer used for analyzing nanoparticles and nanoemulsions during synthesis by measuring their optical properties, such as absorbance at specific wavelengths. It operates on the principle of light absorption, where the intensity of absorbance correlates with the concentration and size of the sample. For nanoparticles, it helps determine key characteristics like size distribution and concentration based on their surface plasmon resonance (SPR) peaks, while for nanoemulsions, it provides insights into droplet size, stability, and homogeneity. NanoDrop's ability to analyze small sample volumes (1-2 µL), its speed, and its non-destructive nature make it a valuable tool for rapid monitoring during nanoparticle or nanoemulsion synthesis, although it may need to be complemented with other techniques (such as DLS or TEM) for full characterization of particle size, distribution, and structure.

CHAPTER 4

MATERIALS AND METHODS

4.1.REVERSE TRANSCRIPTASE-POLYMERASE CHAIN REACTION : COVID-19 ANALYSIS

4.1.1.MATERIALS:

TARGET	REPORTER	QUENCHER
E-Gene	FAM	BHQ1
N-Gene	HEX/VIC	BHQ2
RdRP Gene	CYS	BHQ2
Human RNase	P Texas	BHQ2

Table 1 – Fluorophore-Quencher Combination for SARS-CoV-2 RT-PCR Target.

4.1.2.REAGENTS:

- 2X Master Mix
- 20X Primer and Probe mix
- Nuclease Free dH
- Positive Test Control

4.1.3.PROCEDURE:

Prepare the reaction mixture as per the table given below after deciding the number of reaction cycles

Reagents	1 Rxn	25 Rxn	50 Rxn	100 Rxn
2X Master Mix	12.5 µL	312.5 µL	625 µL	1250 µL
20X Primer and Probe Mix	1.25 µL	31.25 µL	62.5 µL	125 µL
Nuclease Free dH_2O	3.25 µL	81.25 µL	162.5 µL	325 µL
Total	17 µL	425 Ml	850 µL	1700 µL

Table 2 – Mixture of reagents for no. of reaction cycles

Add 8 µL of test RNA per reaction

4.1.4.TEST PREPARATION/REACTION SET-UP:

1. Thaw all components of the kit on ice, mix gently using vortex and spindown the contents for 5 sec and use it immediately.
2. Calculate the number of reactions for each experiment including all controls with one excess reaction volume in the reaction cocktail to accommodate pipetting errors. (eg:number of reaction (n) including controls are 10 add 1 extra reaction during the preparation n+1)
3. Prepare the reaction mix in a 1.5/2 mL tube for the calculated number of samples in Master Mix Preparation room.
4. The assay should be run along with positive controls and negative controls.

STEP	TEMP (°C)	TIME	DETECTION	CYCLE
Reverse Transcription	50	15 min	Off	1
	95	2 min	Off	1
PCR and detection	95	15 sec	Off	45
	58	15 sec	On	

Table 3 – Steps in RT-PCR for detection.

4.2.FLUORESCENENCE MICROSCOPY EXPERIMENTS

4.2.1.CRYSTAL VIOLET BIOFILM FORMATION ASSAY

4.2.1.1.MATERIALS REQUIRED:

- 96-well sterile microtiter plate
- Pipettes and tips (Sterile)
- Crystal Violet Solution
- LB medium
- Microplate reader
- Incubator
- 30 % Acetic acid

4.2.1.2.PROCEDURE:

1. **GROWING A BIOFILM:** Grow a bacterial culture(supplemented with appropriate antibiotics and inducers)overnight in a rich medium (i.e. LB)

- Dilute the overnight culture 1:100 into fresh medium for biofilm assays.
- Add 100 µL of the dilution per well in a 96 well dish. For quantitative assays, 4-8 replicate wells are used for each treatment.
- Incubate the microtiter plate for 4-24 hrs at 37°C.

2. **STAINING THE BIOFILM:**

- After incubation, dump out cells by turning the plate over and shaking out the liquid.
- Gently submerge the plate in a small tub of water. Shake out water. Repeat this process a second time. This step helps remove unattached cells and media components that can be stained in the next step and significantly lowers background staining.
- Add 125 µL of a 0.1% solution of crystal violet in water to each well of the microtiter plate. Wear gloves and a lab coat while making the solution. Use caution when weighing out the CV as the powder is hydroscopic and readily stains clothing, skin, etc.

- Incubate the micro-titre plate at room temperature for 10-15 min.
- Rinse the plate 3-4 times with water by submerging in a tub of water as outlined above, shake out and blot vigorously on a stack of paper towels to rid the plate of all excess cells and dye.
- Turn the micro-titre plate upside down and dry for a few hours or overnight.

3. QUANTIFYING THE BIOFILM

- Add 125 µL of 30% acetic acid in water to each well of the micro-titre plate to solubilize the CV.
- Incubate the microtiter plate at room temperature for 10-15 min.
- Transfer 125 µL of the solubilized CV to a new flat-bottomed micro-titre dish.
- Quantify absorbance in a plate reader at 590 nm using 30% acetic acid in water as the blank.

4.2.2.1.MATERIALS:

- SYTO 9 dye, 3.34 mM (Component A), 300 µL solution in DMSO
- Propidium iodide, 20 mM (Component B), 300 µL solution in DMSO
- BacLight mounting oil (Component C), 10 mL, for bacteria immobilized on membranes. The refractive index at 25°C is 1.517 ± 0.003. DO NOT USE FOR IMMERSION OIL.

4.2.2.2.PROCEDURE:

- Inoculate bacterial culture *Escherichia coli* or *Staphylococcus aureus* in nutrient broth medium along with respective antibiotics and incubate 37°C at incubator overnight
- Concentrate 0.5 mL of the bacterial culture by centrifugation at 10,000 × g for 10–15 minutes.
- Remove the supernatant and resuspend the pellet in 2 mL of appropriate buffer (1X PBS).
- Pellet the samples by centrifugation at 10,000 × g for 10–15 minutes.
- Determine the optical density at 670 nm (OD670) of a 3 mL aliquot of the bacterial suspensions in glass or acrylic absorption cuvettes (1 cm path length).

1. **STAINING OF LIVE/DEAD USING FLURO DYES:**

- Add 3 μL of the dye mixture for each mL of the bacterial suspension, the reagent mixture will contribute 0.3% DMSO to the staining solution. Higher DMSO concentrations may adversely affect staining.
- Mix thoroughly and incubate at room temperature in the dark for 15 minutes.
- Trap 5 μL of the stained bacterial suspension between a slide and an 18 mm square coverslip.
- Visualize under fluorescent microscopy.

2. **FLUORESCENCE MEASUREMENT AND DATA ANALYSIS:**

With the excitation wavelength centered at about 485 nm, measure the fluorescence intensity at a wavelength centered at about 530nm (Emission 1: Green) for each well/entire plate. With the excitation wavelength centered at about 485 nm, measure the fluorescence intensity at a wavelength centered at about 630nm (Emission 2: Red) for each well/entire plate. Analyze the data by dividing the fluorescence intensity of the stained bacterial suspensions (F cell) at emission 1 by the fluorescence intensity at emission 2.

Ratio G/R = F cell, em1 / F cell, em2

Plot the Ratio G/R Versus the percentage of live cells in the bacterial suspensions and calculate the drug toxicity.

4.2.3.ORGANELLES FIXATION AND VISUALIZATION

4.2.3.1.MATERIALS REQUIRED:

- Cell culture or tissue samples
- Glass slides and coverslips
- Micropipettes and tips
- Forceps and scissors (for tissue handling)
- Incubator
- Centrifuge (for cell pellets)
- Fixative agents (e.g., formaldehyde, glutaraldehyde)
- Permeabilization agents (e.g., Triton X-100, saponin)
- Blocking agents (e.g., Bovine Serum Albumin (BSA), normal serum)

- Specific dyes or fluorophore-conjugated antibodies (e.g., DAPI, Mitotracker, Alexa Fluor-labeled antibodies)
- Mounting medium (with or without anti-fade properties)

4.2.3.2.REAGENTS REQUIRED:

1. **Fixatives:**
    - 4% Paraformaldehyde (PFA) in PBS
    - 2.5% Glutaraldehyde in PBS (for electron microscopy)
2. **Permeabilization Solutions:**
    - 0.1% Triton X-100 in PBS
    - 0.1% Saponin in PBS
3. **Blocking Solutions:**
    - 1-5% Bovine Serum Albumin (BSA) in PBS
    - 10% Normal Serum in PBS
4. **Dyes/Labels:**
    - DAPI (for nuclear staining)
    - Mitotracker (for mitochondria)
    - Alexa Fluor-conjugated secondary antibodies (for specific proteins)
5. **Buffers:**
    - Phosphate Buffered Saline (PBS)
    - Tris Buffered Saline (TBS)
6. **Mounting Media:**
    - Prolong Gold Antifade Mountant
    - VECTASHIELD Mounting Medium with DAPI

4.2.3.3.PROCEDURE:

1. Sample Preparation:

- **Cells:**

  Grow adherent cells on glass coverslips or culture them in appropriate dishes.

  For suspension cells, collect by centrifugation and resuspend in a minimal volume.

- **Tissues:**

  Cut tissue samples into small pieces ($\leq 1\ cm^3$) for better fixative penetration.

2. Fixation:

- **For Light Microscopy:**
  - ➢ Rinse cells/tissues with PBS.
  - ➢ Fix with 4% PFA in PBS for 10-15 minutes at room temperature.
  - ➢ Wash 3 times with PBS to remove residual fixative.
- **For Electron Microscopy:**
  - ➢ Fix with 2.5% Glutaraldehyde in PBS for 1-2 hours at room temperature or overnight at 4°C.
  - ➢ Post-fix with 1% osmium tetroxide for enhanced membrane contrast.

3. Permeabilization (if required):

- Treat with 0.1% Triton X-100 in PBS for 5-10 minutes at room temperature for light microscopy.
- Alternatively, use 0.1% saponin in PBS to permeabilize plasma membranes while keeping organelle membranes intact.

4. Blocking:

- Incubate samples in blocking solution (e.g., 1-5% BSA or 10% normal serum) for 30-60 minutes at room temperature to prevent non-specific binding.

5. Staining / Labeling:

- **Fluorescent Dyes:**
  - ➢ Incubate with specific dyes (e.g., DAPI for nuclei, Mitotracker for mitochondria) according to the manufacturer's instructions.

- **Immunofluorescence:**

  - ➢ Incubate with primary antibodies specific to target organelles overnight at 4°C or 1-2 hours at room temperature.

- ➢ Wash 3 times with PBS.
- ➢ Incubate with fluorophore-conjugated secondary antibodies for 1 hour at room temperature in the dark.
- ➢ Wash 3 times with PBS.

6. Mounting:

- Place a drop of mounting medium on the slide.
- Gently place the coverslip over the sample, avoiding air bubbles.
- Seal the edges with nail polish if long-term storage is required.

7. Imaging:

- Observe the samples under a fluorescence or confocal microscope.
- Use appropriate filters for the fluorophores used.

4.3.ULTRA-HIGH PERFORMANCE LIQUID CHROMATOGRAPHY : QUANTIFICATION OF CAFFEINE IN BEVERAGES

4.3.1.MATERIALS REQUIRED:

- Pure caffeine
- Methanol
- Water
- 0.45 µm Filter
- UV- Vis detector

4.3.2.PROCEDURE:

1) **Preparation of Standard:**

- Weigh 200 mg of pure caffeine and dissolve it in 500 mL of warm water.
- Now, make the solution to 1 L(1000mL) after cooling down to room temperature.
- Working standards are prepared by serial dilution method and used for calibrations.

2) **Preparation of Sample:**

- 0.5 g of coffee or other beverage is added to 200 mL of distilled water in a flask.

- Place the flask in the water bath at 90°C for about 20 minutes and stir it continuously.
- Following the heating, cool the flask at room temperature.
- Caffeine is extracted either by using water or by using organic solvents like methanol.
- The extracted solution is filtered using a 0.45μm filer to remove any particulates.

3) **Instrumentation:**

- Use of C18 reverse phase column with a diameter of 250 x 4.6 mm.
- Mobile phase: water: Methanol (80:20)
- Flow rate: 1ml/min
- Injection Volume- 20 μL
- Temperature – 35°C

4) **UHPLC Analysis:**

- Set UHPLC column suitable for quantitative analysis of Caffeine equipped with UV detector with a specific wavelength of 254 nm.
- A small amount of sample of 20 μL is injected into the system.
- The sample passes through the column and gets separated based on its interaction with the column's mobile phase.
- As caffeine elutes from the column, it is detected by the UV detector.
- The detector signal is recorded and the concentration of caffeine in the sample is determined using the calibration curve.

5) **Detection and Data Analysis:**

- Set the UV-Vis detector to 200 nm to 400 nm and prepare the calibration curves using known standards.
- The signal recorded by the detector is compared with known references to confirm the presence and concentration of caffeine. Use software tools for peak integration, comparison with standards, and quantification.

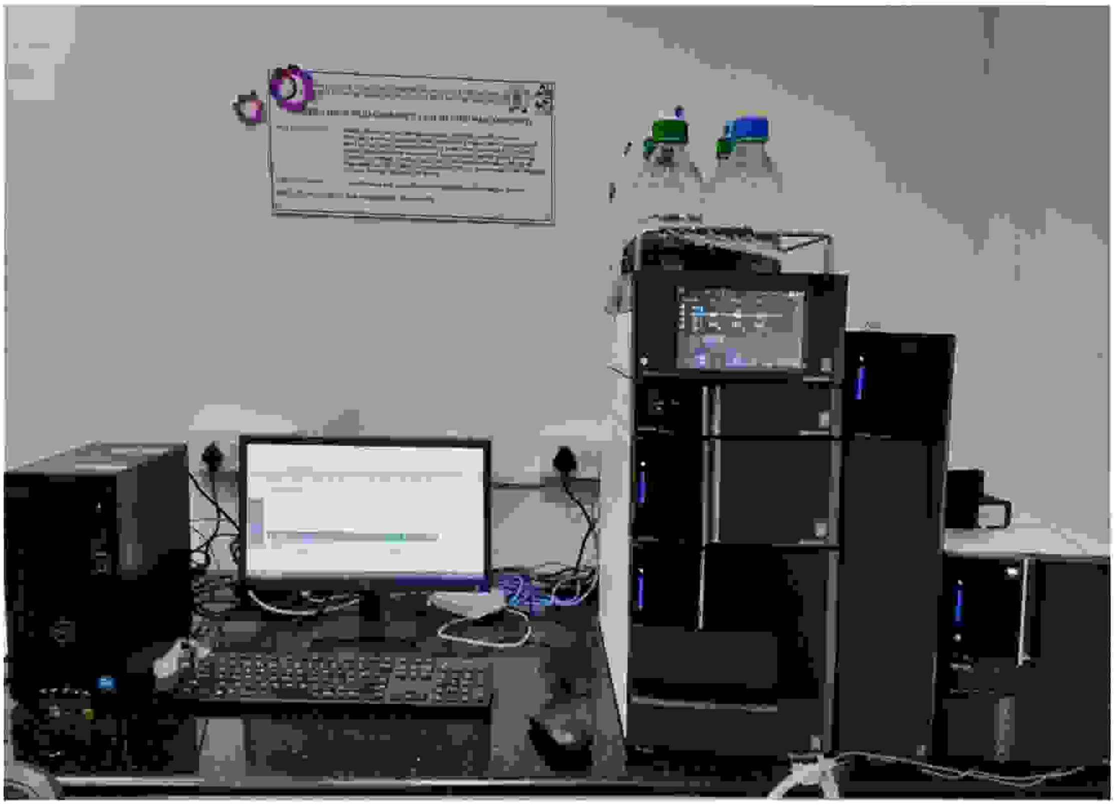


Figure 1 - Ultra-High Performance Liquid Chromatography

4.4.NANOPARTICLES (OR) NANOEMULSION SYNTHESIS : SYNTHESIS OF GREEN NANOPARTICLES

4.4.1.MATERIALS REQUIRED:

- Biological Sources

Plant extracts (leaves, roots, flowers or stem)

Microorganisms (Bacteria, Yeast, Algae, Virus, or Fungi)

- Metal Salts [$AgNO_3$]
- Distilled water
- Centrifuge
- Filter paper
- UV- Visible Spectrophotometer
- Pipettes and tips

4.4.2.PROCEDURE:

a. PREPARATION OF METAL SALT SOLUTION:

- Determine the concentration of the solution required by the experiment
- Calculate the mass of the metal salt.

Mass= Molarity(M) x Molecular weight(g/mol) x Volume(L)

- Add the metal salt to an appropriate volume of distilled water under constant stirring. Desired nanoparticles obtained from the Metal salt solution of $AgNO_3$

b. SYNTHESIS OF GREEN NANOPARTICLES USING PLANT EXTRACTS

- Prepare plant extract by weighing any of the plant parts and wash it with distilled water.
- Grind the plant parts with mortar and pestle, and prepare the extract.
- To the extract, add about 100mL of distilled water and transfer it to a 250mL flask.
- The mixture was boiled for 5 minutes.
- Then cool them at room temperature and filter the extract using filter paper.
- Prepare a metal salt solution for the synthesis of desired metal nanoparticles.
- Add the plant extract to the metal salt solution under continuous stirring
- Centrifuge the solution to collect the nanoparticles.
- Collect the nanoparticles wash them with distilled water and dry them.

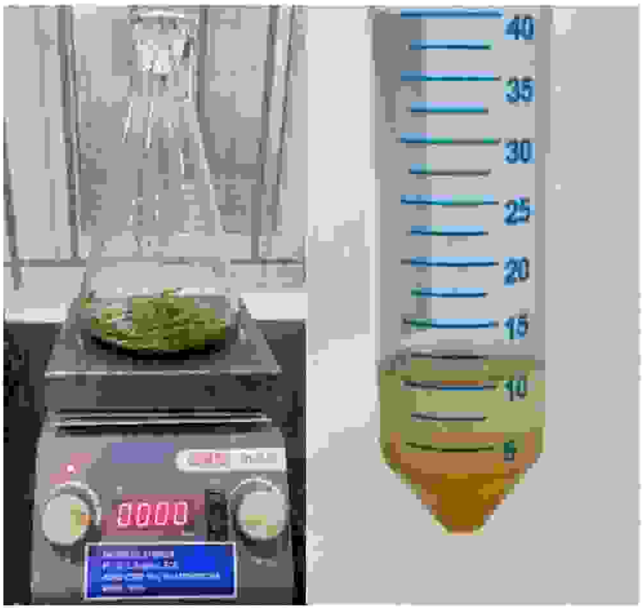


Figure 2 – Plant Extraction and Nano particles from plant extract

CHAPTER 5

RESULTS

5.1.COVID-19 ANALYSIS AND GENE AMPLIFICATION

For all 2019-nCoV genes E, N and RdRP	For RNase P gene	2019-nCoV assay result
Ct < 40	Ct ≤ 40	+
Ct = Undetermined or Ct ≥ 40	Ct ≤ 40	-
Ct = Undetermined or Ct ≥ 40	Ct ≥ 40 Undetermined	Invalid

Table 4 – Determination of Ct value for positive and negative results.

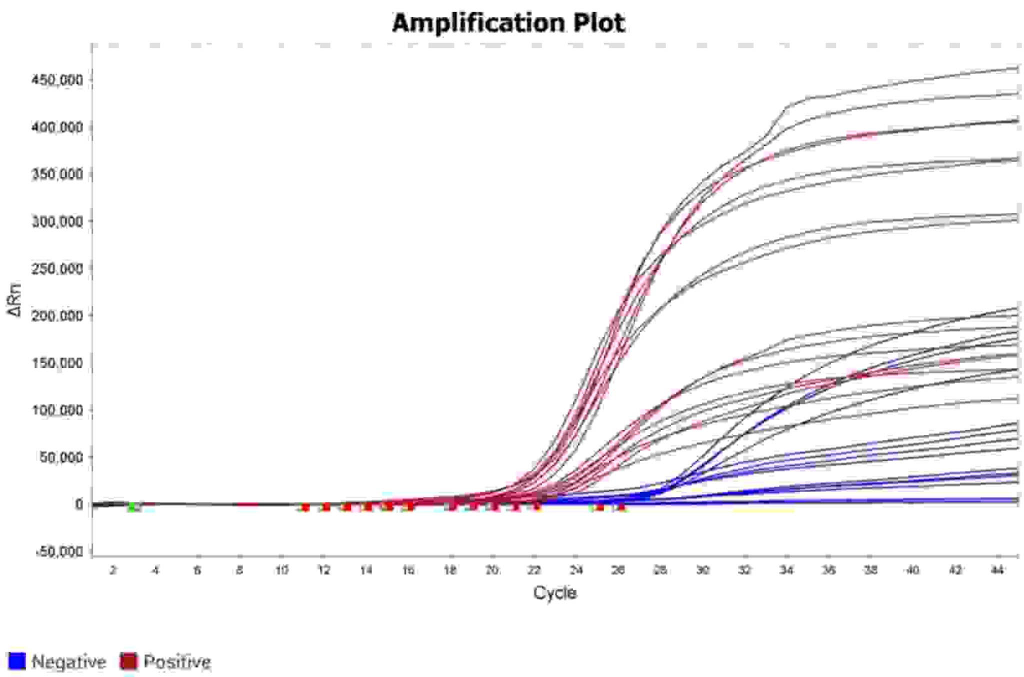


Figure 3 - SARS-CoV-2 in human respiratory specimens using RT-qPCR

DATA INTERPRETATION:

E-Gene FAM	N-Gene HEX/VIC	RdRP CY5	RNaseP TexasRed/ ROX	INTERPRETATION
+	+	+	+/-	2019-nCoV Detected
+	Any 1 gene is +	Any 1 gene is +	+/-	2019-nCoV Detected
-	+	+	+/-	2019-nCoV Detected
-	Any 1 gene is +	Any 1 gene is +	+	2019-nCoV Detected
+	-	-	+	Presumptive positive and sample may be positive for other sarbecovirus
-	-	-	+	2019-nCoV NOT Detected
-	-	-	-	Invalid

Table 5 –Amplification of Covid-19 genes.

5.2. BACTERIAL ATTACHMENT ON THE BIOFILMS BY CRYSTAL VIOLET BIOFILM FORMATION ASSAY

A Cocci shaped bacterial attachment was found on the Biofilms by Crystal Violet Biofilm Formation Assay.

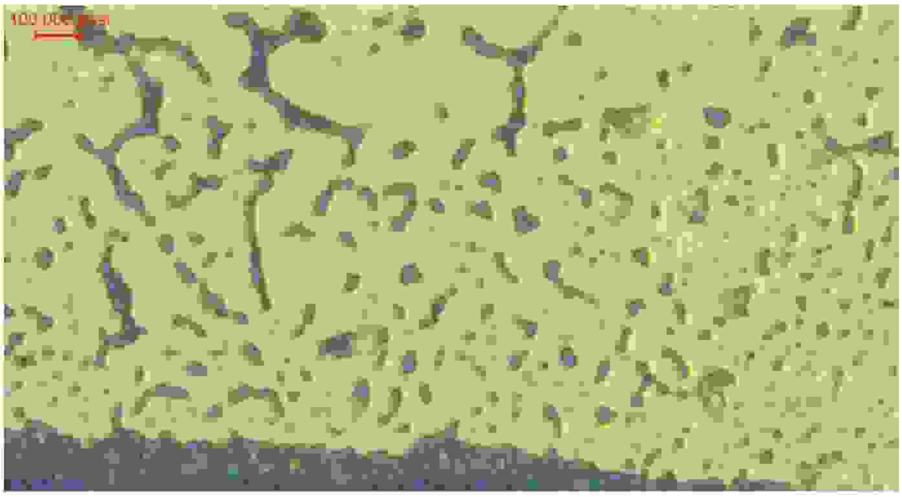


Figure 4 - Stained Bacterial Cells

5.3.DRUG TOXICITY USING FLUORESCENT MICROSCOPY

With the excitation wavelength centered at about 485 nm, measure the fluorescence intensity at a wavelength centered at about 530nm (Emission 1: Green) for each well/entire plate. With the excitation wavelength centered at about 485 nm, measure the fluorescence intensity at a wavelength centered at about 630nm (Emission 2: Red) for each well/entire plate. Analyze the data by dividing the fluorescence intensity of the stained bacterial suspensions (F cell) at emission 1 by the fluorescence intensity at emission 2.

Ratio G/R = F cell, em1 / F cell, em2

Plot the Ratio G/R Versus the percentage of live cells in the bacterial suspensions and calculate the drug toxicity.

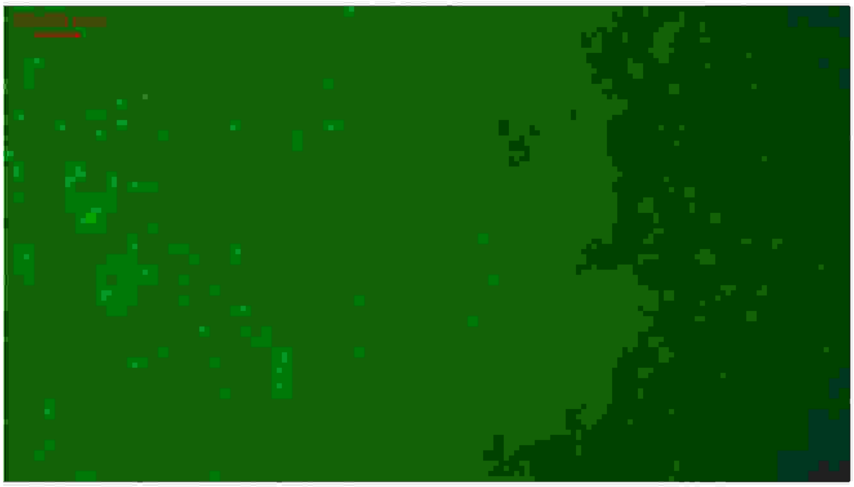


Figure 5 – Live Cells

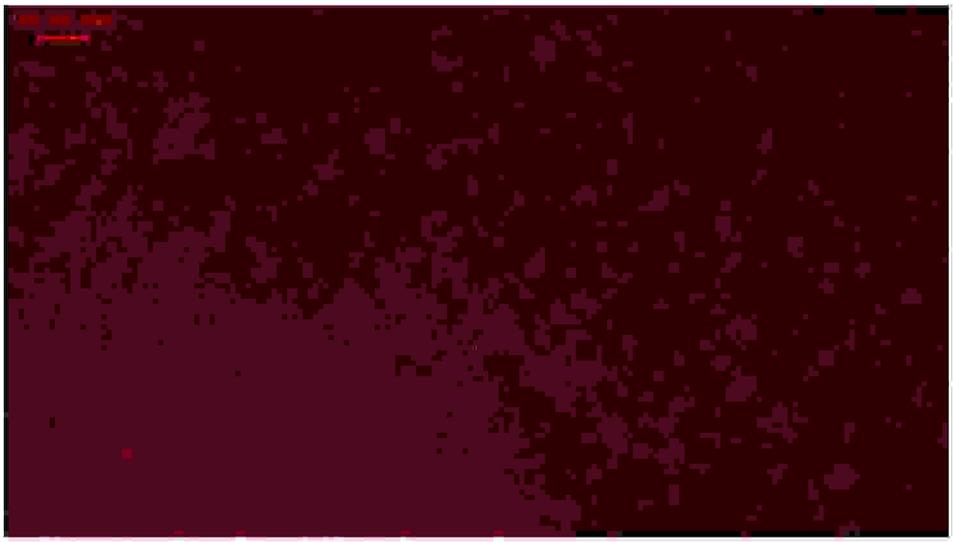


Figure 6 - Dead Cells

The cell viability (LIVE/DEAD) against antibiotics was measured by using fluorescent microscopy.

5.4.ORGANELLES FIXATION AND VISUALIZATION BY DAPI STAINING METHOD

The bacterial cells are observed under the fluorescence microscope by DAPI staining method.

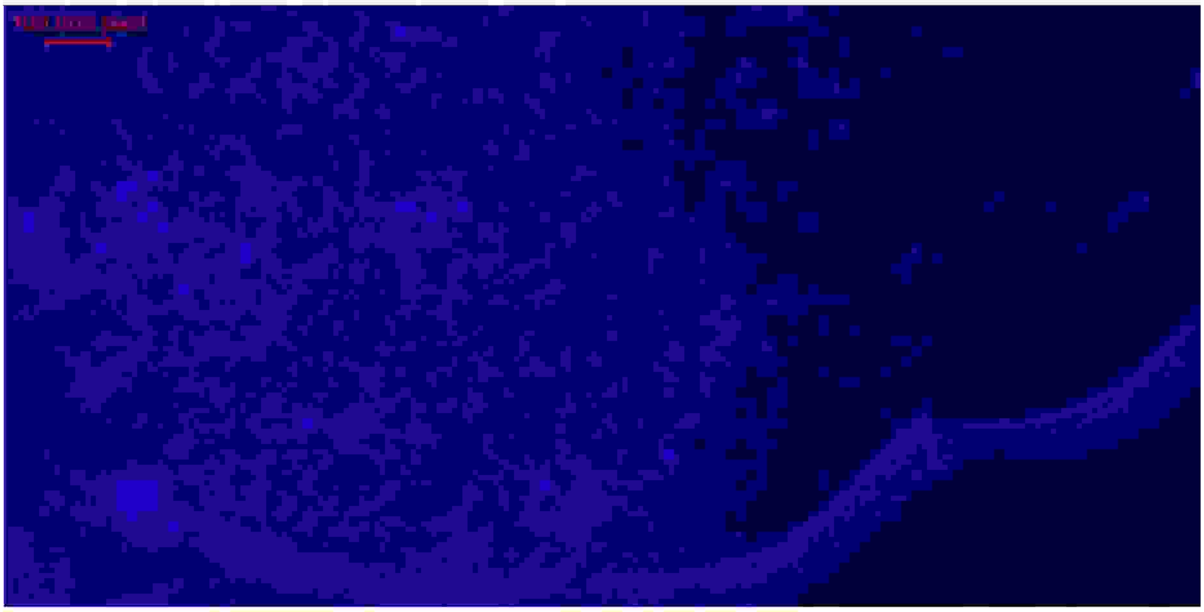


Figure 7 - Stained Bacterial Cells By DAPI Staining Method

5.5.QUANTIFICATION OF CAFFEINE IN BEVERAGES

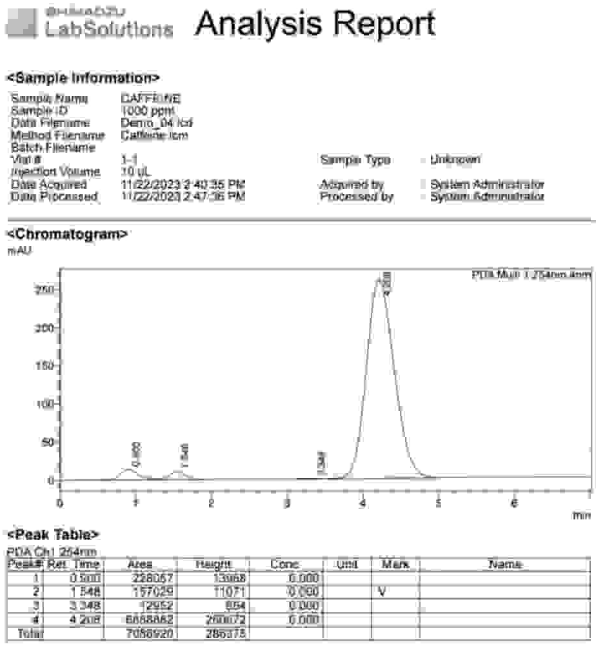
UHPLC Analysis:

- Set UHPLC column suitable for quantitative analysis of Caffeine equipped with UV detector with a specific wavelength of 254 nm.
- A small amount of sample of 20 µL is injected into the system.
- The sample passes through the column and gets separated based on its interaction with the column's mobile phase.
- As caffeine elutes from the column, it is detected by the UV detector.
- The detector signal is recorded and the concentration of caffeine in the sample is determined using the calibration curve.

Detection and Data Analysis:

- Set the UV-Vis detector to 200 nm to 400 nm and prepare the calibration curves using known standards.
- The signal recorded by the detector is compared with known references to confirm the presence and concentration of caffeine. Use software tools for peak integration, comparison with standards, and quantification.

The caffeine was qualitatively analysed from the unknown sample by the Ultra High Performance Liquid Chromatography (UHPLC) technique.



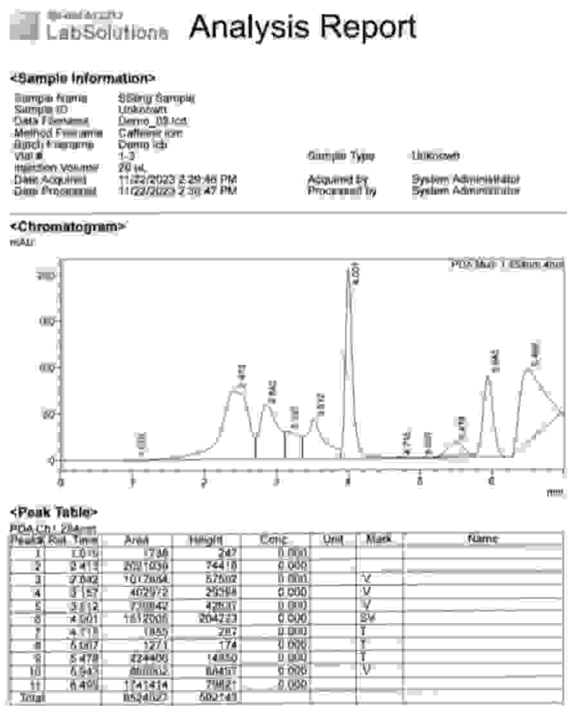


Figure 8 - Caffeine Standard and Unknown Sample

5.6. SYNTHESIS OF GREEN NANOPARTICLES AND CHARACTERIZATION

5.6.1. Characterization of Silver Nanoparticles

- The reduction of pure Ag+ ions was monitored by measuring the absorbance spectrum of the reaction medium using a NanoDrop spectrophotometer.
- The characteristic absorbance peak for silver nanoparticles was identified.

5.6.2.Results and Discussion

1. Visual Observation:

The color change from pale yellow to brown indicated the formation of silver nanoparticles, as silver nanoparticles exhibit a brownish color due to surface plasmon resonance.

2. NanoDrop Spectrophotometry:

The NanoDrop spectrophotometer analysis showed a characteristic peak around 420 nm, confirming the presence of silver nanoparticles. The absorbance peak corresponds to the surface plasmon resonance of silver nanoparticles.

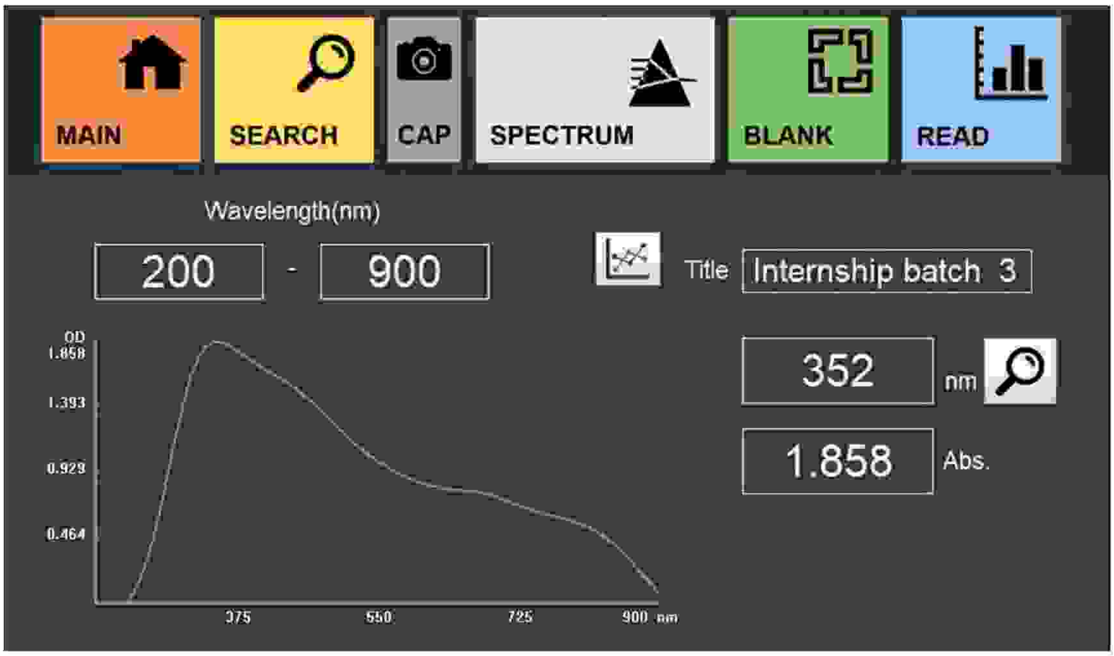


Figure 9 - Absorbance peak at 352 nm

The absorbance peak was observed at 352 nm which infers that silver nanoparticles were synthesised.

CHAPTER 6

CONCLUSION

The industrial training provided a comprehensive platform to integrate theoretical knowledge with practical skills in advanced analytical instrumentation critical to contemporary biological research. Through hands-on experience with Reverse Transcription PCR (RT-PCR), Fluorescence Microscopy, UHPLC, and NanoDrop Spectrophotometry, key competencies were developed in molecular quantification, chromatographic separation, spectrophotometric analysis, and high-resolution cellular imaging. The training not only enhanced instrumental handling and methodological execution but also emphasized analytical thinking, data interpretation, and adherence to standard laboratory practices. Exposure to real-world biological applications—including gene expression profiling, metabolite quantification, and biomolecular visualization—strengthened the understanding of these tools in research and diagnostics.

Overall, the program significantly improved technical proficiency, research aptitude, and confidence in operating precision instruments, forming a strong foundation for future endeavors in biotechnology, molecular biology, and related interdisciplinary fields.

CHAPTER 7

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